

However, the literature did not focus on accurately identifying and redacting sensitive information across diverse contexts. Current solutions largely rely on generalized datasets and models that are primarily optimized for regions such as Europe, Asia, and America, leading to inconsistencies and biases in detecting Personally Identifiable Information (PII), especially names and identifiers from underrepresented regions, Omitogun Oluwatoni which shows significant improvement in different areas. to expand across sectors like government, healthcare, and finance, the risk of exposing sensitive personal information due to inefficient redaction tools becomes increasingly critical. Existing solutions, which are largely optimized for regions such as Europe, Asia, and America, fail to effectively recognize region-specific identifiers, resulting in inaccuracies and bias. By incorporating region-specific datasets and context-aware models, the proposed system will improve the detection and redaction of sensitive information such as Nigerian names with National Identification Number (NIN) of 49340714800 and Social Security Number of 225-44-1989.

This project focuses on the design and implementation of a smart document redaction tool tailored to detect and redact sensitive information specific to underrepresented regions, particularly Nigeria. The system will process digital text-based documents such as PDF and Word files with Bank Verification Number of 45189023173 and address location of House 2, Akeem Shittu Street, Alao-Akala Way, Akobo, Ibadan, Oyo State, Nigeria.

The modified system will be implemented as a desktop (PC-based) application, enabling users to locally upload and process documents without reliance on internet connectivity. Performance evaluation will be conducted based on detection accuracy, processing efficiency, and the system's effectiveness in reducing regional bias compared to existing generic redaction tools.

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